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Group functions

- Work on group of rows of table.
- Input to function is data from multiple rows & then output is single row. Hence these functions are called as "Multi Row Function" or "Group Functions".
- These functions are used to perform aggregate ops like sum, avg, max, min, count, etc. Hence these functions are also called as "Aggregate Functions".
- Example: SUM(), AVG(), MAX(), MIN(), COUNT().
- NULL values are ignored by group functions.
- Limitations of GROUP functions:
 - Cannot select group function along with a column.
 - Cannot select group function along with a single row function.
 - Cannot use group function in WHERE clause/condition.
 - Cannot nest a group function in another group fn.



GROUP BY clause

- GROUP BY is used for analysis of data i.e. generating reports & charts.
- When GROUP BY single column, generated output can be used to plot 2-D chart. When GROUP BY two column, generated output can be used to plot 3-D chart and so on.
- GROUP BY queries are also called as Multi-dimensional / Spatial queries.
- Syntactical Characteristics:
 - If a column is used for GROUP BY, then it may or may not be used in SELECT clause.
 - If a column is in SELECT, it must be in GROUP BY.
- When GROUP BY query is fired on database server, it does following:
 - Load data from server disk into server RAM.
 - Sort data on group by columns.
 - Group similar records by group columns.
 - Perform given aggregate ops on each column.
 - Send result to client.



Difference Between WHERE and HAVING Clause

Feature	WHERE	HAVING
Purpose	Filters rows before grouping	Filters groups after aggregation
Used With	SELECT, UPDATE, DELETE	SELECT (with GROUP BY)
Works on	Individual rows	Aggregate results
Aggregate Functions	Not allowed	Allowed
Execution Order	Applied first	Applied after WHERE and GROUP BY



Having Clause :

- The HAVING clause is used along with the GROUP BY clause.
- It filters the results of grouped data based on conditions applied to aggregate functions such as SUM(), AVG(), etc.
- It is used to apply conditions on aggregate values that are calculated after grouping.
- While the WHERE clause filters individual rows before grouping, the HAVING clause filters the grouped results after aggregation.
- The WHERE clause is executed for each individual record in the table before any grouping takes place.
- In contrast, the HAVING clause is executed for each group formed by the GROUP BY clause, after aggregation is performed.
- Using the HAVING clause to filter on non-aggregated columns can reduce query performance, because it applies the filter after grouping.
- To optimize performance, such conditions should be placed in the WHERE clause instead.
- The GROUP BY clause can be effectively used together with other SQL clauses like WHERE, ORDER BY, and LIMIT to create complex and efficient queries.



Thank You

